

Follow the Gradient

VLC = $\iiint$ Vision
Learning & Control

The power of differentiation

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- The big idea: optimisation by following gradients
- Recap: what are gradients and how do we find them?
- Jacobians as local linear maps
- Jacobian-vector and vector-Jacobian products

The big idea: optimisation by following gradients

- Fundamentally, we're interested in machines that we train by optimising parameters
 - How do we select those parameters?
- In deep learning/differentiable programming we typically define an objective function that we *minimise* (or *maximise*) with respect to those parameters
- This implies that we're looking for points at which the gradient of the objective function is zero w.r.t the parameters

The big idea: optimisation by following gradients

- Gradient based optimisation is a *big* field!
 - First order methods, second order methods, subgradient methods...
- With deep learning we're primarily interested in first-order methods¹.
 - Primarily using variants of gradient descent: a function $F(\mathbf{x})$ has a minima² (or a saddle-point) at a point $\mathbf{x} = \mathbf{a}$ where $\mathbf{a}$ is given by applying $\mathbf{a}_{n+1} = \mathbf{a}_n - \alpha \nabla F(\mathbf{a}_n)$ until convergence from some initial point $\mathbf{a}_0$.

¹Second order gradient optimisers are potentially better, but for systems with many variables are currently impractical as they require computing the Hessian.

²not necessarily global or unique

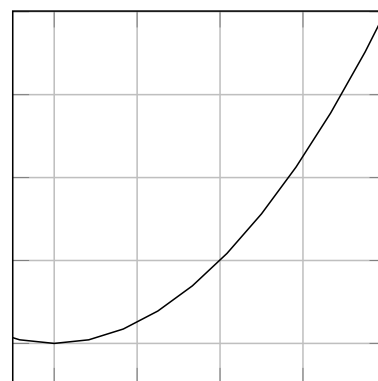
Recap: what are gradients and how do we find them?

The derivative in 1D

- Recall that the gradient of a straight line is $\frac{\Delta y}{\Delta x}$.

- For an arbitrary real-valued function, $f(a)$, we can approximate the derivative, $f'(a)$ using the gradient of the *secant line* defined by $(a, f(a))$ and a point a small distance, h , away $(a + h, f(a + h))$: $f'(a) \approx \frac{f(a+h)-f(a)}{h}$.

- This expression is 'Newton's Quotient' or 'Fermat's Difference Quotient'.
- As h becomes smaller, the approximated derivative becomes more accurate.
- If we take the limit as $h \rightarrow 0$, then we have an exact expression for the derivative:
$$\frac{df}{da} = f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}.$$



Recap: what are gradients and how do we find them?

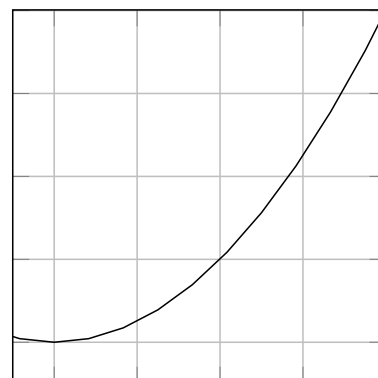
The derivative of $y = x^2$ from first principles

$$\begin{aligned}y &= x^2 \\ \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{x^2 + h^2 + 2hx - x^2}{h} \\ \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{h^2 + 2hx}{h} \\ \frac{dy}{dx} &= \lim_{h \rightarrow 0} (h + 2x) \\ \frac{dy}{dx} &= 2x\end{aligned}$$

Recap: what are gradients and how do we find them?

Aside: numerical approximation of the derivative

- For numerical computation of derivatives it is better to use a “centralised” definition of the derivative:
 - $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a-h)}{2h}$
 - The bit inside the limit is known as the *symmetric difference quotient*
 - For small values of h this has less error than the standard one-sided difference quotient.



Recap: what are gradients and how do we find them?

Aside: numerical approximation of the derivative

- If you are going to use difference quotients to estimate derivatives you need to be aware of potential rounding errors due to floating point representations.
 - Calculating derivatives this way using less than 64-bit precision is rarely going to be useful. (Numbers are not represented exactly, so even if h is represented exactly, $x + h$ will probably not be)
 - You need to pick an appropriate h - too small and the subtraction will have a large rounding error!

Recap: what are gradients and how do we find them?

Derivatives of deeper functions

- Deep learning is all about optimising deeper functions; functions that are compositions of other functions
 - e.g. $z = f \circ g(x) = f(g(x))$
- The chain rule of calculus tells us how to differentiate compositions of functions:
 - $\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}$

Recap: what are gradients and how do we find them?

Example: differentiating $z = x^4$

Note that this is a silly example that just serves to demonstrate the principle!

$$z = x^4$$

$$z = (x^2)^2 = y^2 \quad \text{where} \quad y = x^2$$

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx} = (2y)(2x) = (2x^2)(2x) = 4x^3$$

Equivalently, from first principles:

$$z = x^4$$

$$\frac{dz}{dx} = \lim_{h \rightarrow 0} \frac{(x+h)^4 - x^4}{h}$$

$$\frac{dz}{dx} = \lim_{h \rightarrow 0} \frac{h^4 + 4h^3x + 6h^2x^2 + 4hx^3 + x^4 - x^4}{h}$$

$$\frac{dz}{dx} = \lim_{h \rightarrow 0} h^3 + 4h^2x + 6hx^2 + 4x^3 = 4x^3$$

Recap: what are gradients and how do we find them?

Vector functions

- What if we're dealing with a *vector* function, $\mathbf{y}(t)$?
 - This can be split into its constituent coordinate functions:
 $\mathbf{y}(t) = (y_1(t), \dots, y_n(t))$.
 - Thus the derivative is a vector (the 'tangent vector'),
 $\mathbf{y}'(t) = (y_1'(t), \dots, y_n'(t))$, which consists of the derivatives of the coordinate functions.
 - Equivalently, $\mathbf{y}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{y}(t+h) - \mathbf{y}(t)}{h}$ if the limit exists.

Recap: what are gradients and how do we find them?

Functions of multiple variables: partial differentiation

- What if the function we're trying to deal with has multiple variables³ (e.g. $f(x, y) = x^2 + xy + y^2$)?
 - This expression has a pair of *partial derivatives*, $\frac{\partial f}{\partial x} = 2x + y$ and $\frac{\partial f}{\partial y} = x + 2y$, computed by differentiating with respect to each variable x and y whilst holding the other(s) constant.
- In general, the partial derivative of a function $f(x_1, \dots, x_n)$ at a point $(a_1, \dots, a_n)$ is given by:
$$\frac{\partial f}{\partial x_i}(a_1, \dots, a_n) = \lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_i + h, \dots, a_n) - f(a_1, \dots, a_i, \dots, a_n)}{h}.$$
- The vector of partial derivatives of a scalar-value multivariate function, $f(x_1, \dots, x_n)$ at a point $(a_1, \dots, a_n)$, can be arranged into a vector: $\nabla f(a_1, \dots, a_n) = (\frac{\partial f}{\partial x_1}(a_1, \dots, a_n), \dots, \frac{\partial f}{\partial x_n}(a_1, \dots, a_n))$.
 - This is the **gradient** of f at a .
- In the case of a vector-valued multivariate function, the partial derivatives form a matrix called the **Jacobian**.

³A multivariate function

Recap: what are gradients and how do we find them?

Functions of vectors and matrices: partial differentiation

- For the kinds of functions (and programs) that we'll look at *optimising* in this course have a number of typical properties:
 - They are scalar-valued
 - We'll look at programs with *multiple losses*, but ultimately we can just consider optimising with respect to the *sum* of the losses.
 - They involve multiple variables, which are often wrapped up in the form of vectors or matrices, and more generally tensors.
 - **How will we find the gradients of these?**

Recap: what are gradients and how do we find them?

The chain rule for vectors

Suppose that $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{y} \in \mathbb{R}^n$, g maps from $\mathbb{R}^m$ to $\mathbb{R}^n$ and f maps from $\mathbb{R}^n$ to $\mathbb{R}$.

If $\mathbf{y} = g(\mathbf{x})$ and $z = f(\mathbf{y})$, then

$$\frac{\partial z}{\partial x_i} = \sum_j \frac{\partial z}{\partial y_j} \frac{\partial y_j}{\partial x_i}.$$

Equivalently, in vector notation:

$$\nabla_{\mathbf{x}} z = \left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right)^\top \nabla_{\mathbf{y}} z$$

where $\frac{\partial \mathbf{y}}{\partial \mathbf{x}}$ is the $n \times m$ Jacobian matrix of g .

Recap: what are gradients and how do we find them?

Example: $\nabla_{\mathbf{W}} f(\mathbf{X}\mathbf{W})$

- Let $\mathbf{D} = \mathbf{X}\mathbf{W}$ where the rows of $\mathbf{X} \in \mathbb{R}^{n \times m}$ contain some fixed *features*, and $\mathbf{W} \in \mathbb{R}^{m \times h}$ is a matrix of weights.
- Also let $\mathcal{L} = f(\mathbf{D})$ be some scalar function of $\mathbf{D}$ that we wish to minimise.
- What are the derivatives of $\mathcal{L}$ with respect to the weights $\mathbf{W}$?

Recap: what are gradients and how do we find them?

Example: $\nabla_{\mathbf{W}} f(\mathbf{X}\mathbf{W})$

- Start by considering a specific weight, W_{uv} : $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_{i,j} \frac{\partial \mathcal{L}}{\partial D_{ij}} \frac{\partial D_{ij}}{\partial W_{uv}}$.
- We know that $\frac{\partial D_{ij}}{\partial W_{uv}} = 0$ if $j \neq v$ because D_{ij} is the dot product of row i of $\mathbf{X}$ and column j of $\mathbf{W}$.
- Therefore, we can simplify the summation to only consider cases where $j = v$: $\sum_{i,j} \frac{\partial \mathcal{L}}{\partial D_{ij}} \frac{\partial D_{ij}}{\partial W_{uv}} = \sum_i \frac{\partial \mathcal{L}}{\partial D_{iv}} \frac{\partial D_{iv}}{\partial W_{uv}}$.
- What is $\frac{\partial D_{iv}}{\partial W_{uv}}$?

$$\begin{aligned} D_{iv} &= \sum_{k=1}^m X_{ik} W_{kv} \\ \frac{\partial D_{iv}}{\partial W_{uv}} &= \frac{\partial}{\partial W_{uv}} \sum_{k=1}^m X_{ik} W_{kv} = \sum_{k=1}^m \frac{\partial}{\partial W_{uv}} X_{ik} W_{kv} \\ \therefore \frac{\partial D_{iv}}{\partial W_{uv}} &= X_{iu} \end{aligned}$$

Recap: what are gradients and how do we find them?

Example: $\nabla_{\mathbf{W}} f(\mathbf{X}\mathbf{W})$

- Putting everything together, we have: $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_i \frac{\partial \mathcal{L}}{\partial D_{iv}} X_{iu}$.
- As we're summing over multiplications of scalars, we can change the order: $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_i X_{iu} \frac{\partial \mathcal{L}}{\partial D_{iv}}$.
- and note that the sum over i is doing a dot product with row u and column v if we transpose X_{iu} to $X_{ui}^\top$: $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_i X_{ui}^\top \frac{\partial \mathcal{L}}{\partial D_{iv}}$.
- We can then see that if we want this for all values of $\mathbf{W}$ it simply generalises to: $\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \mathbf{X}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{D}}$.

Recap: what are gradients and how do we find them?

That worked, but the indices were a pain!

- What did we actually do in the previous derivation?
 - We worked out which elements of $\mathbf{D}$ depended on a particular element W_{uv} .
 - We multiplied each local derivative by the corresponding sensitivity $\frac{\partial \mathcal{L}}{\partial D_{ij}}$.
 - We summed over all of the ways in which changing W_{uv} could affect the loss.
- This is all perfectly manageable for $\mathbf{D} = \mathbf{XW}$...
 - ...but we don't want to repeat that index-heavy argument for every operation in a deep model!

Recap: what are gradients and how do we find them?

STOP! What does a gradient actually mean?

- In your early calculus lessons you likely had it hammered into you that gradients represent rates of change of functions.
- This is of course totally true...
- But, it isn't a particularly useful way to think about the gradients of a loss with respect to the weights of a parameterised function.
 - **The gradient of the loss with respect to a parameter tells you how much the loss will change with a small perturbation to that parameter.**

Jacobians as local linear maps

Packaging up all of those partial derivatives

- For a function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$, the Jacobian collects all of the partial derivatives into an $n \times m$ matrix:

$$(J_f(\mathbf{x}))_{ji} = \frac{\partial f_j}{\partial x_i}.$$

- More usefully, the Jacobian describes the best linear approximation to f close to $\mathbf{x}$ (c.f. the first-order Taylor Series expansion):

$$f(\mathbf{x} + \Delta\mathbf{x}) \approx f(\mathbf{x}) + J_f(\mathbf{x})\Delta\mathbf{x}.$$

- So, the Jacobian tells us how a small input perturbation produces a small output perturbation.

Jacobians as local linear maps

Do we actually want the whole Jacobian?

- Suppose f has millions of inputs and millions of outputs.
 - Writing down J_f explicitly would require trillions of entries!
- Fortunately, most of the time we only need to answer one of two questions:
 - 1 Given a particular perturbation of the inputs, what happens to the outputs?
 - 2 Given a particular sensitivity at the outputs, what are the corresponding sensitivities at the inputs?
- These are *Jacobian-vector products* and *vector-Jacobian products*.

Jacobian-vector products

Push a perturbation forwards

- Let $\mathbf{v} \in \mathbb{R}^m$ describe a direction in which we perturb the input to f .
- The corresponding first-order change in the output is the **Jacobian-vector product** (JVP):

$$\begin{aligned}\text{JVP}_f(\mathbf{x}, \mathbf{v}) &= J_f(\mathbf{x})\mathbf{v} \\ &= \left. \frac{d}{d\epsilon} f(\mathbf{x} + \epsilon\mathbf{v}) \right|_{\epsilon=0}.\end{aligned}$$

- A JVP maps one chosen input perturbation to its resulting output perturbation.
 - We can compute this product without ever constructing the complete Jacobian.

Vector-Jacobian products

Pull a sensitivity backwards

- Now let $\mathbf{u} \in \mathbb{R}^n$ describe how sensitive some later scalar value is to the outputs of f .
- The literal vector-Jacobian product is $\mathbf{u}^\top J_f(\mathbf{x})$. As we write sensitivities as column vectors, we use its transpose:

$$\text{VJP}_f(\mathbf{x}, \mathbf{u}) = J_f(\mathbf{x})^\top \mathbf{u}.$$

- Equivalently, this is the gradient of the scalar-valued function $\mathbf{u}^\top f(\mathbf{x})$:

$$J_f(\mathbf{x})^\top \mathbf{u} = \nabla_{\mathbf{x}}(\mathbf{u}^\top f(\mathbf{x})).$$

- A VJP takes sensitivities at the output and propagates them back to the input.

JVPs and VJPs

Returning to $\mathbf{D} = \mathbf{X}\mathbf{W}$

- Consider the function $g(\mathbf{W}) = \mathbf{X}\mathbf{W}$.
- If we perturb the weights by $\Delta\mathbf{W}$, the JVP pushes that perturbation forwards:

$$\Delta\mathbf{D} = \text{JVP}_g(\mathbf{W}, \Delta\mathbf{W}) = \mathbf{X}\Delta\mathbf{W}.$$

- If $\mathbf{G} = \frac{\partial\mathcal{L}}{\partial\mathbf{D}}$ contains the sensitivities arriving from later in the computation, the VJP pulls them backwards:

$$\text{VJP}_g(\mathbf{W}, \mathbf{G}) = \mathbf{X}^\top \mathbf{G} = \frac{\partial\mathcal{L}}{\partial\mathbf{W}}.$$

- This is exactly the result we obtained from the element-wise derivation.

JVPs and VJPs

The chain rule is composition of products

- If $h = g \circ f$, then a perturbation can be pushed forwards one operation at a time:

$$J_h(\mathbf{x})\mathbf{v} = J_g(f(\mathbf{x}))(J_f(\mathbf{x})\mathbf{v}).$$

- Likewise, a sensitivity can be pulled backwards one operation at a time:

$$J_h(\mathbf{x})^\top \mathbf{u} = J_f(\mathbf{x})^\top (J_g(f(\mathbf{x}))^\top \mathbf{u}).$$

- For a scalar loss, the final sensitivity is simply 1.
 - Backpropagation (in the coming lectures) is a sequence of VJPs, starting at the loss and working backwards through the computation.

JVPs and VJPs

What about matrices and higher-order tensors?

- Conceptually, we could flatten every tensor into a vector, construct the corresponding Jacobian, perform the product and reshape the result.
- In practice, we preserve the natural tensor shapes and directly compute the required product.
 - We avoid explicitly constructing a potentially enormous Jacobian.
 - The result has the natural shape of the input for a VJP, or the output for a JVP.
- Later, we'll see how forward- and reverse-mode automatic differentiation compute these products through an arbitrary program.